

线性变换

Linear Transformations

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Introduction

- The connection between linear functions and matrices is at the heart of our subject.
- It is now time to formalize the connections between matrices, vector spaces, and linear functions defined on vector spaces.

Linear Transformations

Let $\mathcal{U}$ and $\mathcal{V}$ be vector spaces over a field $\mathcal{F}$ ($\mathbb{R}$ or $\mathbb{C}$ for us).

- A **linear transformation** from $\mathcal{U}$ into $\mathcal{V}$ is defined to be a linear function $\mathbf{T}$ mapping $\mathcal{U}$ into $\mathcal{V}$. That is,

$$\mathbf{T}(\mathbf{x} + \mathbf{y}) = \mathbf{T}(\mathbf{x}) + \mathbf{T}(\mathbf{y}) \quad \text{and} \quad \mathbf{T}(\alpha\mathbf{x}) = \alpha\mathbf{T}(\mathbf{x})$$

or, equivalently,

$$\mathbf{T}(\alpha\mathbf{x} + \mathbf{y}) = \alpha\mathbf{T}(\mathbf{x}) + \mathbf{T}(\mathbf{y}) \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathcal{U}, \alpha \in \mathcal{F}.$$

- A **linear operator** on $\mathcal{U}$ is defined to be a linear transformation from $\mathcal{U}$ into itself—i.e., a linear function mapping $\mathcal{U}$ back into $\mathcal{U}$.

Some Example

- The function $\mathbf{0}(x) = \mathbf{0}$ that maps all vectors in a space $\mathcal{U}$ to the zero vector in another space $\mathcal{V}$ is a linear transformation from $\mathcal{U}$ into $\mathcal{V}$, and, not surprisingly, it is called the **zero transformation**.
- The function $\mathbf{I}(x) = x$ that maps every vector from a space $\mathcal{U}$ back to itself is a linear operator on $\mathcal{U}$. $\mathbf{I}$ is called the **identity operator** on $\mathcal{U}$.
- For $\mathbf{A} \in \Re^{m \times n}$ and $x \in \Re^{n \times 1}$, the function $\mathbf{T}(x) = \mathbf{A}x$ is a linear transformation from $\Re^n$ into $\Re^m$ because matrix multiplication satisfies $\mathbf{A}(\alpha x + y) = \alpha \mathbf{A}x + \mathbf{A}y$. $\mathbf{T}$ is a linear operator on $\Re^n$ if A is $n \times n$.
- If $\mathcal{W}$ is the vector space of all functions from $\Re$ to $\Re$, and if $\mathcal{V}$ is the space of all differentiable functions from $\Re$ to $\Re$, then the mapping $\mathbf{D}(f) = df/dx$ is a linear transformation from $\mathcal{V}$ into $\mathcal{W}$ because

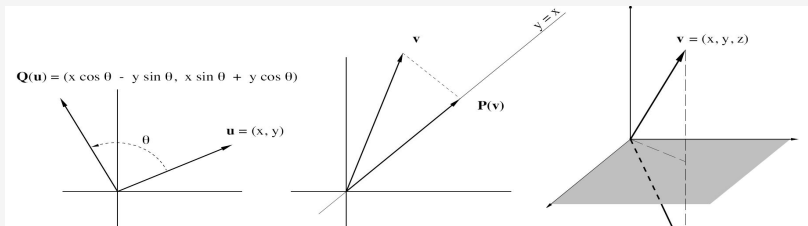
$$\frac{d(\alpha f + g)}{dx} = \alpha \frac{df}{dx} + \frac{dg}{dx}.$$

- If $\mathcal{V}$ is the space of all continuous functions from $\mathfrak{R}$ into $\mathfrak{R}$, then the mapping defined by $\mathbf{T}(f) = \int_0^x f(t)dt$ is a linear operator on $\mathcal{V}$ because

$$\int_0^x [\alpha f(t) + g(t)]dt = \alpha \int_0^x f(t)dt + \int_0^x g(t)dt.$$

- The rotator $\mathbf{Q}$ that rotates vectors $\mathbf{u}$ in $\mathfrak{R}^2$ counterclockwise through an angle θ , is a linear operator on $\mathfrak{R}^2$ because the “action” of $\mathbf{Q}$ on $\mathbf{u}$ can be described by matrix multiplication in the sense that the coordinates of the rotated vector $\mathbf{Q}(\mathbf{u})$ are given by

$$\mathbf{Q}(\mathbf{u}) = \begin{pmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$



- The projector $\mathbf{P}$ that maps $\mathbf{v} = (x, y, z) \in \mathfrak{R}^3$ to its orthogonal projection $(x, y, 0)$ in the xy -plane, is a linear operator on $\mathfrak{R}^3$ because if $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$, then

$$\mathbf{P}(\alpha\mathbf{u} + \mathbf{v}) = (\alpha u_1 + v_1, \alpha u_2 + v_2, 0) = \alpha\mathbf{P}(\mathbf{u}) + \mathbf{P}(\mathbf{v}).$$

- The reflector $\mathbf{R}$ that maps each vector $v = (x, y, z) \in \mathfrak{R}^3$ to its reflection $\mathbf{R}(v) = (x, y, -z)$ about the xy -plane is a linear operator on $\mathfrak{R}^3$.
- Can all linear transformations be represented by matrices?
- Linear transformations on finite-dimensional spaces will always have matrix representations.
- To see why, the concept of coordinates in higher dimensions must first be understood.
- Recall that if $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ is a basis for a vector space $\mathcal{U}$, then $\mathbf{v} = \alpha_1\mathbf{u}_1 + \alpha_2\mathbf{u}_2 + \dots + \alpha_n\mathbf{u}_n$.
- α_i 's are uniquely determined by $\mathbf{v}$, which are called the **coordinates of $\mathbf{v}$ with respect to $\mathcal{B}$** . Denote $[\mathbf{v}]_{\mathcal{B}} = (\alpha_1, \alpha_2, \dots, \alpha_n)^T$.

- From now on, $\mathcal{S} = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ will denote the standard basis of unit vectors for $\mathfrak{R}^n$.
- Linear transformations possess coordinates in the same way vectors do because linear transformations from $\mathcal{U}$ to $\mathcal{V}$ also form a vector space.

Space of Linear Transformations

- For each pair of vector spaces $\mathcal{U}$ and $\mathcal{V}$ over $\mathcal{F}$, the set $\mathcal{L}(\mathcal{U}, \mathcal{V})$ of all linear transformations from $\mathcal{U}$ to $\mathcal{V}$ is a vector space over $\mathcal{F}$.
- Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ and $\mathcal{B}' = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be bases for $\mathcal{U}$ and $\mathcal{V}$, respectively, and let $\mathbf{B}_{ji}$ be the linear transformation from $\mathcal{U}$ into $\mathcal{V}$ defined by $\mathbf{B}_{ji}(\mathbf{u}) = \xi_j \mathbf{v}_i$, where $(\xi_1, \xi_2, \dots, \xi_n)^T = [\mathbf{u}]_{\mathcal{B}}$. That is, pick off the j^{th} coordinate of $\mathbf{u}$, and attach it to $\mathbf{v}_i$.
 - ▷ $\mathcal{B}_{\mathcal{L}} = \{\mathbf{B}_{ji}\}_{j=1 \dots m}^{i=1 \dots n}$ is a basis for $\mathcal{L}(\mathcal{U}, \mathcal{V})$.
 - ▷ $\dim \mathcal{L}(\mathcal{U}, \mathcal{V}) = (\dim \mathcal{U}) (\dim \mathcal{V})$.

- Prove $\mathcal{B}_{\mathcal{L}}$ is a basis by demonstrating that it is a linearly independent spanning set for $\mathcal{L}(\mathcal{U}, \mathcal{V})$. To establish linear independence,

suppose $\sum_{j,i} \eta_{ji} \mathbf{B}_{ji} = \mathbf{0}$ for scalars η_{ji} , and observe that for each $\mathbf{u}_k \in \mathcal{B}$,

$$\mathbf{B}_{ji}(\mathbf{u}_k) = \begin{cases} \mathbf{v}_i & \text{if } j = k \\ \mathbf{0} & \text{if } j \neq k \end{cases} \implies \mathbf{0} = \left(\sum_{j,i} \eta_{ji} \mathbf{B}_{ji} \right)(\mathbf{u}_k) = \sum_{j,i} \eta_{ji} \mathbf{B}_{ji}(\mathbf{u}_k) = \sum_{i=1}^m \eta_{ki} \mathbf{v}_i.$$

For each k , the independence of $\mathcal{B}'$ implies that $\eta_{ki} = 0$ for each i , and thus $\mathcal{B}_{\mathcal{L}}$ is linearly independent. To see that $\mathcal{B}_{\mathcal{L}}$ spans $\mathcal{L}(\mathcal{U}, \mathcal{V})$, let $\mathbf{T} \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, and determine the action of $\mathbf{T}$ on any $\mathbf{u} \in \mathcal{U}$ by using $\mathbf{u} = \sum_{j=1}^n \xi_j \mathbf{u}_j$ and $\mathbf{T}(\mathbf{u}_j) = \sum_{i=1}^m \alpha_{ij} \mathbf{v}_i$ to write

$$\begin{aligned} \mathbf{T}(\mathbf{u}) &= \mathbf{T}\left(\sum_{j=1}^n \xi_j \mathbf{u}_j\right) = \sum_{j=1}^n \xi_j \mathbf{T}(\mathbf{u}_j) = \sum_{j=1}^n \xi_j \sum_{i=1}^m \alpha_{ij} \mathbf{v}_i \\ &= \sum_{i,j} \alpha_{ij} \xi_j \mathbf{v}_i = \sum_{i,j} \alpha_{ij} \mathbf{B}_{ji}(\mathbf{u}). \end{aligned}$$

This holds for all $\mathbf{u} \in \mathcal{U}$, so $\mathbf{T} = \sum_{i,j} \alpha_{ij} \mathbf{B}_{ji}$, and thus $\mathcal{B}_{\mathcal{L}}$ spans $\mathcal{L}(\mathcal{U}, \mathcal{V})$.

Coordinate Matrix Representations

Let $\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ and $\mathcal{B}' = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be bases for $\mathcal{U}$ and $\mathcal{V}$, respectively. The *coordinate matrix* of $\mathbf{T} \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ with respect to the pair $(\mathcal{B}, \mathcal{B}')$ is defined to be the $m \times n$ matrix

$$[\mathbf{T}]_{\mathcal{B}\mathcal{B}'} = \left([\mathbf{T}(\mathbf{u}_1)]_{\mathcal{B}'} \mid [\mathbf{T}(\mathbf{u}_2)]_{\mathcal{B}'} \mid \cdots \mid [\mathbf{T}(\mathbf{u}_n)]_{\mathcal{B}'} \right).$$

In other words, if $\mathbf{T}(\mathbf{u}_j) = \alpha_{1j}\mathbf{v}_1 + \alpha_{2j}\mathbf{v}_2 + \cdots + \alpha_{mj}\mathbf{v}_m$, then

$$[\mathbf{T}(\mathbf{u}_j)]_{\mathcal{B}'} = \begin{pmatrix} \alpha_{1j} \\ \alpha_{2j} \\ \vdots \\ \alpha_{mj} \end{pmatrix} \text{ and } [\mathbf{T}]_{\mathcal{B}\mathcal{B}'} = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \cdots & \alpha_{mn} \end{pmatrix}.$$

When $\mathbf{T}$ is a linear operator on $\mathcal{U}$, and when there is only one basis involved, $[\mathbf{T}]_{\mathcal{B}}$ is used in place of $[\mathbf{T}]_{\mathcal{B}\mathcal{B}}$ to denote the (necessarily square) coordinate matrix of $\mathbf{T}$ with respect to $\mathcal{B}$.

- **Example:** If $\mathbf{P}$ is the projector, determine the coordinate matrix $[\mathbf{P}]_{\mathfrak{B}}$ with respect to the basis

$$\mathcal{B} = \left\{ \mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \mathbf{u}_3 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\}.$$

- **Solution:** According to the above results, the j^{th} column in $[\mathbf{P}]_{\mathfrak{B}}$ is $[\mathbf{P}(\mathbf{u}_j)]_{\mathfrak{B}}$. Therefore,

$$\mathbf{P}(\mathbf{u}_1) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 1\mathbf{u}_1 + 1\mathbf{u}_2 - 1\mathbf{u}_3 \implies [\mathbf{P}(\mathbf{u}_1)]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix},$$

$$\mathbf{P}(\mathbf{u}_2) = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = 0\mathbf{u}_1 + 3\mathbf{u}_2 - 2\mathbf{u}_3 \implies [\mathbf{P}(\mathbf{u}_2)]_{\mathcal{B}} = \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix},$$

$$\mathbf{P}(\mathbf{u}_3) = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = 0\mathbf{u}_1 + 3\mathbf{u}_2 - 2\mathbf{u}_3 \implies [\mathbf{P}(\mathbf{u}_3)]_{\mathcal{B}} = \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix},$$

so that $[\mathbf{P}]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 3 & 3 \\ -1 & -2 & -2 \end{pmatrix}.$

- At the heart of linear algebra is the realization that the theory of finite dimensional linear transformations is essentially the same as the theory of matrices.
- This is due primarily to the fundamental fact that the action of a linear transformation $\mathbf{T}$ on a vector $\mathbf{u}$ is precisely matrix multiplication between the coordinates of $\mathbf{T}$ and the coordinates of $\mathbf{u}$.

Action as Matrix Multiplication

Let $\mathbf{T} \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, and let $\mathcal{B}$ and $\mathcal{B}'$ be bases for $\mathcal{U}$ and $\mathcal{V}$, respectively. For each $\mathbf{u} \in \mathcal{U}$, the action of $\mathbf{T}$ on $\mathbf{u}$ is given by matrix multiplication between their coordinates in the sense that

$$[\mathbf{T}(\mathbf{u})]_{\mathcal{B}'} = [\mathbf{T}]_{\mathcal{B}\mathcal{B}'} [\mathbf{u}]_{\mathcal{B}}.$$

- **Example:** Show how the action of the operator $\mathbf{D}(p(t)) = dp/dt$ on the space $\mathcal{P}_3$ of polynomials of degree three or less is given by matrix multiplication.

Solution: The coordinate matrix of $\mathbf{D}$ with respect to the basis $\mathcal{B} = \{1, t, t^2, t^3\}$ is

$$[\mathbf{D}]_{\mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

If $\mathbf{p} = p(t) = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \alpha_3 t^3$, then $\mathbf{D}(\mathbf{p}) = \alpha_1 + 2\alpha_2 t + 3\alpha_3 t^2$ so that

$$[\mathbf{p}]_{\mathcal{B}} = \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} \quad \text{and} \quad [\mathbf{D}(\mathbf{p})]_{\mathcal{B}} = \begin{pmatrix} \alpha_1 \\ 2\alpha_2 \\ 3\alpha_3 \\ 0 \end{pmatrix}.$$

The action of $\mathbf{D}$ is accomplished by means of matrix multiplication because

$$[\mathbf{D}(\mathbf{p})]_{\mathcal{B}} = \begin{pmatrix} \alpha_1 \\ 2\alpha_2 \\ 3\alpha_3 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = [\mathbf{D}]_{\mathcal{B}} [\mathbf{p}]_{\mathcal{B}}.$$

- For $\mathbf{T} \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $\mathbf{L} \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, the composition of $\mathbf{L}$ with $\mathbf{T}$ is defined to be the function $\mathbf{C} : \mathcal{U} \rightarrow \mathcal{W}$ such that $\mathbf{C}(x) = \mathbf{L}(\mathbf{T}(x))$.
- This composition denoted by $\mathbf{C}(x) = \mathbf{LT}$, is also a linear transformation because

$$\begin{aligned}\mathbf{C}(\alpha \mathbf{x} + \mathbf{y}) &= \mathbf{L}(\mathbf{T}(\alpha \mathbf{x} + \mathbf{y})) = \mathbf{L}(\alpha \mathbf{T}(\mathbf{x}) + \mathbf{T}(\mathbf{y})) \\ &= \alpha \mathbf{L}(\mathbf{T}(\mathbf{x})) + \mathbf{L}(\mathbf{T}(\mathbf{y})) = \alpha \mathbf{C}(\mathbf{x}) + \mathbf{C}(\mathbf{y}).\end{aligned}$$

- If $\mathfrak{B}$, $\mathfrak{B}'$ and $\mathfrak{B}''$ are bases for $\mathcal{U}$, $\mathcal{V}$ and $\mathcal{W}$, respectively, then $\mathbf{C}$ must have a coordinate matrix representation with respect to $(\mathfrak{B}, \mathfrak{B}'')$.
- So it's only natural to ask how $[\mathbf{C}]_{\mathfrak{B}\mathfrak{B}''}$ is related to $[\mathbf{L}]_{\mathfrak{B}'\mathfrak{B}''}$ and $[\mathbf{T}]_{\mathfrak{B}\mathfrak{B}'}$:

$$[\mathbf{C}]_{\mathfrak{B}\mathfrak{B}''} = [\mathbf{L}]_{\mathfrak{B}'\mathfrak{B}''} [\mathbf{T}]_{\mathfrak{B}\mathfrak{B}'}.$$

Connections with Matrix Algebra

- If $\mathbf{T}, \mathbf{L} \in \mathcal{L}(\mathcal{U}, \mathcal{V})$, and if $\mathcal{B}$ and $\mathcal{B}'$ are bases for $\mathcal{U}$ and $\mathcal{V}$, then
 - ▷ $[\alpha \mathbf{T}]_{\mathcal{B}\mathcal{B}'} = \alpha [\mathbf{T}]_{\mathcal{B}\mathcal{B}'}$ for scalars α ,
 - ▷ $[\mathbf{T} + \mathbf{L}]_{\mathcal{B}\mathcal{B}'} = [\mathbf{T}]_{\mathcal{B}\mathcal{B}'} + [\mathbf{L}]_{\mathcal{B}\mathcal{B}'}$.
- If $\mathbf{T} \in \mathcal{L}(\mathcal{U}, \mathcal{V})$ and $\mathbf{L} \in \mathcal{L}(\mathcal{V}, \mathcal{W})$, and if $\mathcal{B}$, $\mathcal{B}'$, and $\mathcal{B}''$ are bases for $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{W}$, respectively, then $\mathbf{LT} \in \mathcal{L}(\mathcal{U}, \mathcal{W})$, and
 - ▷ $[\mathbf{LT}]_{\mathcal{B}\mathcal{B}''} = [\mathbf{L}]_{\mathcal{B}'\mathcal{B}''} [\mathbf{T}]_{\mathcal{B}\mathcal{B}'}$.
- If $\mathbf{T} \in \mathcal{L}(\mathcal{U}, \mathcal{U})$ is invertible in the sense that $\mathbf{T}\mathbf{T}^{-1} = \mathbf{T}^{-1}\mathbf{T} = \mathbf{I}$ for some $\mathbf{T}^{-1} \in \mathcal{L}(\mathcal{U}, \mathcal{U})$, then for every basis $\mathcal{B}$ of $\mathcal{U}$,
 - ▷ $[\mathbf{T}^{-1}]_{\mathcal{B}} = [\mathbf{T}]_{\mathcal{B}}^{-1}$.

- From the composition $\mathbf{C} = \mathbf{LT}$ of the two linear transformations $\mathbf{T} : \mathfrak{R}^3 \rightarrow \mathfrak{R}^2$ and $\mathbf{L} : \mathfrak{R}^2 \rightarrow \mathfrak{R}^2$ defined by

$$\mathbf{T}(x, y, z) = (x + y, y - z) \text{ and } \mathbf{L}(u, v) = (2u - v, u),$$

using the standard bases for $\mathfrak{R}^2$ and $\mathfrak{R}^3$ to verify the above results.

Change of Basis and Similarity

- By their nature, coordinate matrix representations are basis dependent.
- It's desirable to study linear transformations without reference to particular bases because some bases may force a coordinate matrix representation to exhibit special properties that are not present in the coordinate matrix relative to other bases.
- It's necessary to somehow identify properties of coordinate matrices that are invariant among all bases.
- These are properties intrinsic to the transformation itself.
- The following discussion is limited to a single finite-dimensional space $\mathcal{V}$ and to linear operators on $\mathcal{V}$.
- Begin by examining how the coordinates of $\mathbf{v} \in \mathcal{V}$ change as the basis for $\mathcal{V}$ changes.
- Consider two different bases

$$\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\} \quad \text{and} \quad \mathcal{B}' = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}.$$

- Throughout this section $\mathbf{T}$ will denote the linear operator such that

$$\mathbf{T}(\mathbf{y}_i) = \mathbf{x}_i \quad i = 1, 2, \dots, n.$$

- $\mathbf{T}$ is called the **change of basis operator**.

Changing Vector Coordinates

Let $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ and $\mathcal{B}' = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n\}$ be bases for $\mathcal{V}$, and let $\mathbf{T}$ and $\mathbf{P}$ be the associated change of basis operator and change of basis matrix, respectively—i.e., $\mathbf{T}(\mathbf{y}_i) = \mathbf{x}_i$, for each i , and

$$\mathbf{P} = [\mathbf{T}]_{\mathcal{B}} = [\mathbf{T}]_{\mathcal{B}'} = [\mathbf{I}]_{\mathcal{B}\mathcal{B}'} = \left([\mathbf{x}_1]_{\mathcal{B}'} \mid [\mathbf{x}_2]_{\mathcal{B}'} \mid \cdots \mid [\mathbf{x}_n]_{\mathcal{B}'} \right).$$

- $[\mathbf{v}]_{\mathcal{B}'} = \mathbf{P}[\mathbf{v}]_{\mathcal{B}}$ for all $\mathbf{v} \in \mathcal{V}$.
- $\mathbf{P}$ is nonsingular.

- To see this, observe that

$$\mathbf{x}_i = \sum_{j=1}^n \alpha_j \mathbf{y}_j \implies \mathbf{T}(\mathbf{x}_i) = \sum_{j=1}^n \alpha_j \mathbf{T}(\mathbf{y}_j) = \sum_{j=1}^n \alpha_j \mathbf{x}_j,$$

which means $[\mathbf{x}_i]_{\mathcal{B}'} = [\mathbf{T}(\mathbf{x}_i)]_{\mathcal{B}}$, so,

$$[\mathbf{T}]_{\mathcal{B}} = \begin{pmatrix} [\mathbf{T}(\mathbf{x}_1)]_{\mathcal{B}} & [\mathbf{T}(\mathbf{x}_2)]_{\mathcal{B}} & \cdots & [\mathbf{T}(\mathbf{x}_n)]_{\mathcal{B}} \end{pmatrix} = \begin{pmatrix} [\mathbf{x}_1]_{\mathcal{B}'} & [\mathbf{x}_2]_{\mathcal{B}'} & \cdots & [\mathbf{x}_n]_{\mathcal{B}'} \end{pmatrix} = [\mathbf{T}]_{\mathcal{B}'}.$$

The fact that $[\mathbf{I}]_{\mathcal{B}\mathcal{B}'} = [\mathbf{T}]_{\mathcal{B}}$ follows because $[\mathbf{I}(\mathbf{x}_i)]_{\mathcal{B}'} = [\mathbf{x}_i]_{\mathcal{B}'}$. The matrix

$$\mathbf{P} = [\mathbf{I}]_{\mathcal{B}\mathcal{B}'} = [\mathbf{T}]_{\mathcal{B}} = [\mathbf{T}]_{\mathcal{B}'}$$

- $[\mathbf{v}]_{\mathcal{B}'} = [\mathbf{I}]_{\mathcal{B}\mathcal{B}'}[\mathbf{v}]_{\mathcal{B}} = \mathbf{P}[\mathbf{v}]_{\mathcal{B}}$.
- $\mathbf{P}$ is nonsingular because $\mathbf{T}$ is invertible ($\mathbf{T}^{-1}(\mathbf{x}_i) = \mathbf{y}_i$).
- $[\mathbf{T}^{-1}]_{\mathcal{B}} = [\mathbf{T}]_{\mathcal{B}}^{-1} = \mathbf{P}^{-1}$.

- The change of basis operator $\mathbf{T}$ acts as

$$\mathbf{T}(\text{new basis}) = \text{old basis}.$$

- While the change of basis matrix $\mathbf{P}$ acts as

$$\text{new coordinates} = \mathbf{P}(\text{old coordinates}).$$

- **Example:** For the space $\mathcal{P}_2$ of polynomials of degree 2 or less, determine the change of basis matrix $\mathbf{P}$ from $\mathcal{B}$ to $\mathcal{B}'$, where

$$\mathcal{B} = \{1, t, t^2\} \quad \text{and} \quad \mathcal{B}' = \{1, 1+t, 1+t+t^2\},$$

and then find the coordinates of $q(t) = 3 + 2t + 4t^2$ relative to $\mathcal{B}'$.

- **Solution:** The change of basis matrix from $\mathcal{B}$ to $\mathcal{B}'$ is

$$\mathbf{P} = ([x_1]_{\mathcal{B}'} \mid [x_2]_{\mathcal{B}'} \mid [x_3]_{\mathcal{B}'}).$$

In this case, $\mathbf{x}_1 = 1$, $\mathbf{x}_2 = t$, and $\mathbf{x}_3 = t^2$, and $\mathbf{y}_1 = 1$, $\mathbf{y}_2 = 1 + t$, and $\mathbf{y}_3 = 1 + t + t^2$, so the coordinates $[\mathbf{x}_i]_{\mathcal{B}'}$ are computed as follows:

$$\begin{aligned} 1 &= 1(1) + 0(1+t) + 0(1+t+t^2) = 1\mathbf{y}_1 + 0\mathbf{y}_2 + 0\mathbf{y}_3, \\ t &= -1(1) + 1(1+t) + 0(1+t+t^2) = -1\mathbf{y}_1 + 1\mathbf{y}_2 + 0\mathbf{y}_3, \\ t^2 &= 0(1) - 1(1+t) + 1(1+t+t^2) = 0\mathbf{y}_1 - 1\mathbf{y}_2 + 1\mathbf{y}_3. \end{aligned}$$

Therefore,

$$\mathbf{P} = \left([\mathbf{x}_1]_{\mathcal{B}'} \mid [\mathbf{x}_2]_{\mathcal{B}'} \mid [\mathbf{x}_3]_{\mathcal{B}'} \right) = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix},$$

and the coordinates of $\mathbf{q} = q(t) = 3 + 2t + 4t^2$ with respect to $\mathcal{B}'$ are

$$[\mathbf{q}]_{\mathcal{B}'} = \mathbf{P}[\mathbf{q}]_{\mathcal{B}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}.$$

To independently check that these coordinates are correct, simply verify that

$$q(t) = 1(1) - 2(1+t) + 4(1+t+t^2).$$

- It's now rather easy to describe how the coordinate matrix of a linear operator changes as the underlying basis changes.

Changing Matrix Coordinates

Let $\mathbf{A}$ be a linear operator on $\mathcal{V}$, and let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for $\mathcal{V}$. The coordinate matrices $[\mathbf{A}]_{\mathcal{B}}$ and $[\mathbf{A}]_{\mathcal{B}'}$ are related as follows.

$$[\mathbf{A}]_{\mathcal{B}} = \mathbf{P}^{-1}[\mathbf{A}]_{\mathcal{B}'}\mathbf{P}, \quad \text{where} \quad \mathbf{P} = [\mathbf{I}]_{\mathcal{B}\mathcal{B}'}$$

is the change of basis matrix from $\mathcal{B}$ to $\mathcal{B}'$. Equivalently,

$$[\mathbf{A}]_{\mathcal{B}'} = \mathbf{Q}^{-1}[\mathbf{A}]_{\mathcal{B}}\mathbf{Q}, \quad \text{where} \quad \mathbf{Q} = [\mathbf{I}]_{\mathcal{B}'\mathcal{B}} = \mathbf{P}^{-1}$$

is the change of basis matrix from $\mathcal{B}'$ to $\mathcal{B}$.

Problem: Consider the linear operator $\mathbf{A}(x, y) = (y, -2x + 3y)$ on $\mathbb{R}^2$ along with the two bases

$$\mathcal{S} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \quad \text{and} \quad \mathcal{S}' = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}.$$

First compute the coordinate matrix $[\mathbf{A}]_{\mathcal{S}}$ as well as the change of basis matrix $\mathbf{Q}$ from $\mathcal{S}'$ to $\mathcal{S}$, and then use these two matrices to determine $[\mathbf{A}]_{\mathcal{S}'}$.

Solution: The matrix of $\mathbf{A}$ relative to $\mathcal{S}$ is obtained by computing

$$\mathbf{A}(\mathbf{e}_1) = \mathbf{A}(1, 0) = (0, -2) = (0)\mathbf{e}_1 + (-2)\mathbf{e}_2,$$

$$\mathbf{A}(\mathbf{e}_2) = \mathbf{A}(0, 1) = (1, 3) = (1)\mathbf{e}_1 + (3)\mathbf{e}_2,$$

so that $[\mathbf{A}]_{\mathcal{S}} = ([\mathbf{A}(\mathbf{e}_1)]_{\mathcal{S}} \mid [\mathbf{A}(\mathbf{e}_2)]_{\mathcal{S}}) = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}.$

The change of basis matrix from $\mathcal{S}'$ to $\mathcal{S}$ is

$$\mathbf{Q} = ([\mathbf{y}_1]_{\mathcal{S}} \mid [\mathbf{y}_2]_{\mathcal{S}}) = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

and the matrix of $\mathbf{A}$ with respect to $\mathcal{S}'$ is

$$[\mathbf{A}]_{\mathcal{S}'} = \mathbf{Q}^{-1}[\mathbf{A}]_{\mathcal{S}}\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Similarity

- Matrices $\mathbf{B}_{n \times n}$ and $\mathbf{C}_{n \times n}$ are said to be *similar matrices* whenever there exists a nonsingular matrix $\mathbf{Q}$ such that $\mathbf{B} = \mathbf{Q}^{-1}\mathbf{C}\mathbf{Q}$. We write $\mathbf{B} \simeq \mathbf{C}$ to denote that $\mathbf{B}$ and $\mathbf{C}$ are similar.
- The linear operator $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ defined by $f(\mathbf{C}) = \mathbf{Q}^{-1}\mathbf{C}\mathbf{Q}$ is called a *similarity transformation*.

- Any two coordinate matrices of a given linear operator must be similar.
- Must any two similar matrices be coordinate matrices of the same linear operator?
- Similar matrices represent the same linear operator.**
- So the coordinate-independent properties are exactly the ones that are **similarity invariant** (invariant under similarity transformations).
- Determining and studying similarity invariants is an important part of linear algebra and matrix theory.

Invariant Subspaces

For a linear operator $\mathbf{T}$ on a vector space $\mathcal{V}$, and for $\mathcal{X} \subseteq \mathcal{V}$,

$$\mathbf{T}(\mathcal{X}) = \{\mathbf{T}(\mathbf{x}) \mid \mathbf{x} \in \mathcal{X}\}$$

is the set of all possible images of vectors from $\mathcal{X}$ under the transformation $\mathbf{T}$. Notice that $\mathbf{T}(\mathcal{V}) = R(\mathbf{T})$. When $\mathcal{X}$ is a subspace of $\mathcal{V}$, it follows that $\mathbf{T}(\mathcal{X})$ is also a subspace of $\mathcal{V}$, but $\mathbf{T}(\mathcal{X})$ is usually not related to $\mathcal{X}$. However, in some special cases it can happen that $\mathbf{T}(\mathcal{X}) \subseteq \mathcal{X}$, and such subspaces are the focus of this section.

Invariant Subspaces

- For a linear operator $\mathbf{T}$ on $\mathcal{V}$, a subspace $\mathcal{X} \subseteq \mathcal{V}$ is said to be an *invariant subspace* under $\mathbf{T}$ whenever $\mathbf{T}(\mathcal{X}) \subseteq \mathcal{X}$.
- In such a situation, $\mathbf{T}$ can be considered as a linear operator on $\mathcal{X}$ by forgetting about everything else in $\mathcal{V}$ and restricting $\mathbf{T}$ to act only on vectors from $\mathcal{X}$. Hereafter, this *restricted operator* will be denoted by $\mathbf{T}_{/\mathcal{X}}$.

Problem: For

$$\mathbf{A} = \begin{pmatrix} 4 & 4 & 4 \\ -2 & -2 & -5 \\ 1 & 2 & 5 \end{pmatrix}, \quad \mathbf{x}_1 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{x}_2 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix},$$

show that the subspace $\mathcal{X}$ spanned by $\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2\}$ is an invariant subspace under $\mathbf{A}$. Then describe the restriction $\mathbf{A}|_{\mathcal{X}}$ and determine the coordinate matrix of $\mathbf{A}|_{\mathcal{X}}$ relative to $\mathcal{B}$.

Solution: Observe that $\mathbf{A}\mathbf{x}_1 = 2\mathbf{x}_1 \in \mathcal{X}$ and $\mathbf{A}\mathbf{x}_2 = \mathbf{x}_1 + 2\mathbf{x}_2 \in \mathcal{X}$, so the image of any $\mathbf{x} = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2 \in \mathcal{X}$ is back in $\mathcal{X}$ because

$$\mathbf{A}\mathbf{x} = \mathbf{A}(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = \alpha\mathbf{A}\mathbf{x}_1 + \beta\mathbf{A}\mathbf{x}_2 = 2\alpha\mathbf{x}_1 + \beta(\mathbf{x}_1 + 2\mathbf{x}_2) = (2\alpha + \beta)\mathbf{x}_1 + 2\beta\mathbf{x}_2.$$

This equation completely describes the action of $\mathbf{A}$ restricted to $\mathcal{X}$, so

$$\mathbf{A}|_{\mathcal{X}}(\mathbf{x}) = (2\alpha + \beta)\mathbf{x}_1 + 2\beta\mathbf{x}_2 \quad \text{for each} \quad \mathbf{x} = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2 \in \mathcal{X}.$$

Since $\mathbf{A}|_{\mathcal{X}}(\mathbf{x}_1) = 2\mathbf{x}_1$ and $\mathbf{A}|_{\mathcal{X}}(\mathbf{x}_2) = \mathbf{x}_1 + 2\mathbf{x}_2$, we have

$$\left[\mathbf{A}|_{\mathcal{X}}\right]_{\mathcal{B}} = \left(\left[\mathbf{A}|_{\mathcal{X}}(\mathbf{x}_1)\right]_{\mathcal{B}} \mid \left[\mathbf{A}|_{\mathcal{X}}(\mathbf{x}_2)\right]_{\mathcal{B}} \right) = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

The invariant subspaces for a linear operator $\mathbf{T}$ are important because they produce simplified coordinate matrix representations of $\mathbf{T}$. To understand how this occurs, suppose $\mathcal{X}$ is an invariant subspace under $\mathbf{T}$, and let

$$\mathcal{B}_{\mathcal{X}} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r\}$$

be a basis for $\mathcal{X}$ that is part of a basis

$$\mathcal{B} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r, \mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_q\}$$

for the entire space $\mathcal{V}$. To compute $[\mathbf{T}]_{\mathcal{B}}$, recall from the definition of coordinate matrices that

$$[\mathbf{T}]_{\mathcal{B}} = \left([\mathbf{T}(\mathbf{x}_1)]_{\mathcal{B}} \mid \cdots \mid [\mathbf{T}(\mathbf{x}_r)]_{\mathcal{B}} \mid [\mathbf{T}(\mathbf{y}_1)]_{\mathcal{B}} \mid \cdots \mid [\mathbf{T}(\mathbf{y}_q)]_{\mathcal{B}} \right).$$

Because each $\mathbf{T}(\mathbf{x}_j)$ is contained in $\mathcal{X}$, only the first r vectors from $\mathcal{B}$ are needed to represent each $\mathbf{T}(\mathbf{x}_j)$, so, for $j = 1, 2, \dots, r$,

$$\mathbf{T}(\mathbf{x}_j) = \sum_{i=1}^r \alpha_{ij} \mathbf{x}_i \quad \text{and} \quad [\mathbf{T}(\mathbf{x}_j)]_{\mathcal{B}} = \begin{pmatrix} \alpha_{1j} \\ \vdots \\ \alpha_{rj} \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

The space

$$\mathcal{Y} = \text{span}\{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_q\}$$

may not be an invariant subspace for $\mathbf{T}$, so all the basis vectors in $\mathcal{B}$ may be needed to represent the $\mathbf{T}(\mathbf{y}_j)$'s. Consequently, for $j = 1, 2, \dots, q$,

$$\mathbf{T}(\mathbf{y}_j) = \sum_{i=1}^r \beta_{ij} \mathbf{x}_i + \sum_{i=1}^q \gamma_{ij} \mathbf{y}_i \quad \text{and} \quad [\mathbf{T}(\mathbf{y}_j)]_{\mathcal{B}} = \begin{pmatrix} \beta_{1j} \\ \vdots \\ \beta_{rj} \\ \gamma_{1j} \\ \vdots \\ \gamma_{qj} \end{pmatrix}.$$

$$[\mathbf{T}]_{\mathcal{B}} = \begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1r} & \beta_{11} & \cdots & \beta_{1q} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \alpha_{r1} & \cdots & \alpha_{rr} & \beta_{r1} & \cdots & \beta_{rq} \\ 0 & \cdots & 0 & \gamma_{11} & \cdots & \gamma_{1q} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \gamma_{q1} & \cdots & \gamma_{qq} \end{pmatrix}.$$

- The equation $\mathbf{T}(\mathbf{x}_j) = \sum_{i=1}^r \alpha_{ij} \mathbf{x}_i$ mean that

$$[\mathbf{T}/\mathcal{X}]_{\mathcal{B}_\mathcal{X}} = \begin{pmatrix} \alpha_{11} & \alpha_{12} & \cdots & \alpha_{1r} \\ \alpha_{21} & \alpha_{22} & \cdots & \alpha_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{r1} & \alpha_{r2} & \cdots & \alpha_{rr} \end{pmatrix}$$

- So, we have

$$[\mathbf{T}]_{\mathcal{B}} = \begin{pmatrix} [\mathbf{T}/\mathcal{X}]_{\mathcal{B}_\mathcal{X}} & \mathbf{B}_{r \times q} \\ \mathbf{0} & \mathbf{C}_{q \times q} \end{pmatrix}.$$

- This says that the matrix representation for $\mathbf{T}$ can be made to be block triangular whenever a basis for an invariant subspace is available.
- The more invariant subspaces we can find, the more tools we have to construct simplified matrix representations.

This notion easily generalizes in the sense that if $\mathcal{B} = \mathcal{B}_X \cup \mathcal{B}_Y \cup \cdots \cup \mathcal{B}_Z$ is a basis for $\mathcal{V}$, where $\mathcal{B}_X, \mathcal{B}_Y, \dots, \mathcal{B}_Z$ are bases for invariant subspaces under $\mathbf{T}$ that have dimensions $r_1, r_2, \dots, r_k$, respectively, then $[\mathbf{T}]_{\mathcal{B}}$ has the block-diagonal form

$$[\mathbf{T}]_{\mathcal{B}} = \begin{pmatrix} \mathbf{A}_{r_1 \times r_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{r_2 \times r_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{C}_{r_k \times r_k} \end{pmatrix},$$

where

$$\mathbf{A} = [\mathbf{T}/\mathcal{X}]_{\mathcal{B}_X}, \quad \mathbf{B} = [\mathbf{T}/\mathcal{Y}]_{\mathcal{B}_Y}, \quad \dots, \quad \mathbf{C} = [\mathbf{T}/\mathcal{Z}]_{\mathcal{B}_Z}.$$

The situations discussed above are also reversible in the sense that if the matrix representation of $\mathbf{T}$ has a block-triangular form

$$[\mathbf{T}]_{\mathcal{B}} = \begin{pmatrix} \mathbf{A}_{r \times r} & \mathbf{B}_{r \times q} \\ \mathbf{0} & \mathbf{C}_{q \times q} \end{pmatrix}$$

relative to some basis

$$\mathcal{B} = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_q\},$$

then the r -dimensional subspace $\mathcal{U} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\}$ spanned by the first r vectors in $\mathcal{B}$ must be an invariant subspace under $\mathbf{T}$. Furthermore, if the matrix representation of $\mathbf{T}$ has a block-diagonal form

$$[\mathbf{T}]_{\mathcal{B}} = \begin{pmatrix} \mathbf{A}_{r \times r} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{q \times q} \end{pmatrix}$$

relative to $\mathcal{B}$, then both

$$\mathcal{U} = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_r\} \quad \text{and} \quad \mathcal{W} = \text{span}\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_q\}$$

must be invariant subspaces for $\mathbf{T}$.

The general statement concerning invariant subspaces and coordinate matrix representations is given below.

Invariant Subspaces and Matrix Representations

Let $\mathbf{T}$ be a linear operator on an n -dimensional space $\mathcal{V}$, and let $\mathcal{X}, \mathcal{Y}, \dots, \mathcal{Z}$ be subspaces of $\mathcal{V}$ with respective dimensions $r_1, r_2, \dots, r_k$ and bases $\mathcal{B}_{\mathcal{X}}, \mathcal{B}_{\mathcal{Y}}, \dots, \mathcal{B}_{\mathcal{Z}}$. Furthermore, suppose that $\sum_i r_i = n$ and $\mathcal{B} = \mathcal{B}_{\mathcal{X}} \cup \mathcal{B}_{\mathcal{Y}} \cup \dots \cup \mathcal{B}_{\mathcal{Z}}$ is a basis for $\mathcal{V}$.

- The subspace $\mathcal{X}$ is an invariant subspace under $\mathbf{T}$ if and only if $[\mathbf{T}]_{\mathcal{B}}$ has the block-triangular form

$$[\mathbf{T}]_{\mathcal{B}} = \begin{pmatrix} \mathbf{A}_{r_1 \times r_1} & \mathbf{B} \\ \mathbf{0} & \mathbf{C} \end{pmatrix}, \quad \text{in which case} \quad \mathbf{A} = [\mathbf{T}/_{\mathcal{X}}]_{\mathcal{B}_{\mathcal{X}}}.$$

- The subspaces $\mathcal{X}, \mathcal{Y}, \dots, \mathcal{Z}$ are all invariant under $\mathbf{T}$ if and only if $[\mathbf{T}]_{\mathcal{B}}$ has the block-diagonal form

$$[\mathbf{T}]_{\mathcal{B}} = \begin{pmatrix} \mathbf{A}_{r_1 \times r_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{r_2 \times r_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{C}_{r_k \times r_k} \end{pmatrix},$$

in which case

$$\mathbf{A} = [\mathbf{T}/\mathcal{X}]_{\mathcal{B}_x}, \quad \mathbf{B} = [\mathbf{T}/\mathcal{Y}]_{\mathcal{B}_y}, \quad \dots, \quad \mathbf{C} = [\mathbf{T}/\mathcal{Z}]_{\mathcal{B}_z}.$$

An important corollary concerns the special case in which the linear operator $\mathbf{T}$ is in fact an $n \times n$ matrix and $\mathbf{T}(\mathbf{v}) = \mathbf{T}\mathbf{v}$ is a matrix–vector multiplication.

Triangular and Diagonal Block Forms

When $\mathbf{T}$ is an $n \times n$ matrix, the following two statements are true.

- $\mathbf{Q}$ is a nonsingular matrix such that

$$\mathbf{Q}^{-1}\mathbf{T}\mathbf{Q} = \begin{pmatrix} \mathbf{A}_{r \times r} & \mathbf{B}_{r \times q} \\ \mathbf{0} & \mathbf{C}_{q \times q} \end{pmatrix}$$

if and only if the first r columns in $\mathbf{Q}$ span an invariant subspace under $\mathbf{T}$.

- $\mathbf{Q}$ is a nonsingular matrix such that

$$\mathbf{Q}^{-1}\mathbf{T}\mathbf{Q} = \begin{pmatrix} \mathbf{A}_{r_1 \times r_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_{r_2 \times r_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{C}_{r_k \times r_k} \end{pmatrix}$$

if and only if $\mathbf{Q} = (\mathbf{Q}_1 | \mathbf{Q}_2 | \cdots | \mathbf{Q}_k)$ in which $\mathbf{Q}_i$ is $n \times r_i$, and the columns of each $\mathbf{Q}_i$ span an invariant subspace under $\mathbf{T}$.

Problem: Find all subspaces of $\mathbb{R}^2$ that are invariant under

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}.$$

Solution: The trivial subspace $\{\mathbf{0}\}$ is the only zero-dimensional invariant subspace, and the entire space $\mathbb{R}^2$ is the only two-dimensional invariant subspace. The real problem is to find all one-dimensional invariant subspaces. If $\mathcal{M}$ is a one-dimensional subspace spanned by $\mathbf{x} \neq \mathbf{0}$ such that $\mathbf{A}(\mathcal{M}) \subseteq \mathcal{M}$, then

$$\mathbf{A}\mathbf{x} \in \mathcal{M} \implies \text{there is a scalar } \lambda \text{ such that } \mathbf{A}\mathbf{x} = \lambda\mathbf{x} \implies (\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}.$$

In other words, $\mathcal{M} \subseteq N(\mathbf{A} - \lambda\mathbf{I})$. Since $\dim \mathcal{M} = 1$, it must be the case that $N(\mathbf{A} - \lambda\mathbf{I}) \neq \{\mathbf{0}\}$, and consequently λ must be a scalar such that $(\mathbf{A} - \lambda\mathbf{I})$ is a singular matrix. Row operations produce

$$\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} -\lambda & 1 \\ -2 & 3 - \lambda \end{pmatrix} \longrightarrow \begin{pmatrix} -2 & 3 - \lambda \\ -\lambda & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} -2 & 3 - \lambda \\ 0 & 1 + (\lambda^2 - 3\lambda)/2 \end{pmatrix},$$

and it is clear that $(\mathbf{A} - \lambda\mathbf{I})$ is singular if and only if $1 + (\lambda^2 - 3\lambda)/2 = 0$ —i.e., if and only if λ is a root of

$$\lambda^2 - 3\lambda + 2 = 0.$$

Thus $\lambda = 1$ and $\lambda = 2$, and straightforward computation yields the two one-dimensional invariant subspaces

$$\mathcal{M}_1 = N(\mathbf{A} - \mathbf{I}) = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\} \quad \text{and} \quad \mathcal{M}_2 = N(\mathbf{A} - 2\mathbf{I}) = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}.$$

In passing, notice that $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^2$, and

$$[\mathbf{A}]_{\mathcal{B}} = \mathbf{Q}^{-1} \mathbf{A} \mathbf{Q} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \text{where} \quad \mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

In general, scalars λ for which $(\mathbf{A} - \lambda\mathbf{I})$ is singular are called the **eigenvalues** of $\mathbf{A}$, and the nonzero vectors in $N(\mathbf{A} - \lambda\mathbf{I})$ are known as the associated **eigenvectors** for $\mathbf{A}$. As this example indicates, eigenvalues and eigenvectors are of fundamental importance in identifying invariant subspaces and reducing matrices by means of similarity transformations.

Exercise

- Determine which of the following functions are linear operators on $\mathfrak{R}^2$
 - $\mathbf{T}(x, y) = (x, 1 + y)$,
 - $\mathbf{T}(x, y) = (0, xy)$,
 - $\mathbf{T}(x, y) = (x^2, y^2)$.
- Explain why $\mathbf{T}(0) = 0$ for every linear transformation $\mathbf{T}$.
- Explain why rank is a similarity invariant.
- $\mathbf{A}(x, y, z) = (x + 2y - z, -y, x + 7z)$ is a linear operator on $\mathfrak{R}^3$.
 - Determine $[\mathbf{A}]_{\mathcal{S}}$, where $\mathcal{S}$ is the standard basis.
 - Determine $[\mathbf{A}]_{\mathcal{S}'}$ as well as the nonsingular matrix $\mathbf{Q}$ such that

$$[\mathbf{A}]_{\mathcal{S}'} = \mathbf{Q}^{-1}[\mathbf{A}]_{\mathcal{S}}\mathbf{Q} \quad \text{for} \quad \mathcal{S}' = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

- Let $\mathbf{T}$ be an arbitrary linear operator on a vector space $\mathcal{V}$.
 - Is the trivial subspace $\{\mathbf{o}\}$ invariant under $\mathbf{T}$?
 - Is the entire space $\mathcal{V}$ invariant under $\mathbf{T}$?

6. Describe all of the subspace that are invariant under the identity operator $\mathbf{I}$ on a space $\mathcal{V}$.
7. Let $\mathbf{T}$ be the linear operator on $\mathfrak{R}^4$ defined by

$$\mathbf{T}(x_1, x_2, x_3, x_4) = (x_1 + x_2 + 2x_3 - x_4, x_2 + x_4, 2x_3 - x_4, x_3 + x_4),$$

and let $\mathcal{X} = \text{span}\{e_1, e_2\}$ be the subspace that is spanned by the first two unit vectors in $\mathfrak{R}^4$.

- (a) Explain why $\mathcal{X}$ is invariant under $\mathbf{T}$.
- (b) Determine $[\mathbf{T}/\mathcal{X}]_{\{e_1, e_2\}}$.
- (c) Describe the structure of $[\mathbf{T}]_{\mathcal{B}}$, where $\mathcal{B}$ is any basis obtained from an extension of $\{e_1, e_2\}$.